IMU-Based Step and Orientation Detection

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***Abstract*—This review compares six approaches to detecting steps and heading from inertial sensor data for pedestrian dead reckoning, and identifies those suited to a device with only an accelerometer and a magnetometer.**

Keywords—pedestrian dead reckoning, step detection, heading estimation, accelerometer, magnetometer

I. Introduction

Pedestrian dead reckoning (PDR) adds a displacement to the previous position each time a step is detected [1], [2]. It avoids double integration of acceleration, whose error grows quickly unless a foot-mounted sensor allows zero-velocity updates [1], [3]. Step and heading errors still accumulate, and heading is repeatedly reported as the dominant one [3]–[5]. This review compares six works with a view to a wearable device that has only an accelerometer and a magnetometer.

II. Step Detection

The detectors differ first in the signal examined. Three works rotate acceleration into a world frame and use the vertical component, making detection independent of device posture [1], [4], [5]. Jiménez et al.[3] and Brajdic and Harle [6] use the acceleration magnitude instead, the latter because the vertical axis is hard to isolate on a loosely carried phone. Tian et al. classify how the phone is carried (held, swinging or in a pocket) and select one axis per mode, since no single threshold on the magnitude fits all modes [2].

Two works threshold the local variance of acceleration: on a foot-mounted sensor this missed 1 of 955 steps [3], and in a trouser pocket it gave under 1% count error, though only for the steps of one leg [4]. The others detect peaks in low-pass filtered acceleration. Jin et al. require a maximum and minimum to differ by a calibrated threshold and be at least 150 ms apart [1]; Kang and Han test peak height, peak-to-valley difference and slope [5]; Tian et al. use a fixed threshold, a 0.2 s minimum step interval and a false-peak rejection window, detecting over 98% of steps [2]. Comparing ten algorithms on 130 walks by 27 people with six phone placements, Brajdic and Harle found all had median errors under 3%, and windowed peak detection matched hidden Markov and wavelet methods at 1.3% while being much simpler [6]. Steps are hardest to detect at the start and end of a walk [3], [6]. Tian et al. filter at 5 Hz because step frequency lies between 0.5 and 5 Hz [2], and the best settings in [6] were a 0.31 s moving average with one peak per 0.59 s window. Simple detectors also fire on non-walking movements such as handling the device, so false steps must be rejected [6]. Fixed thresholds are fragile because signal amplitude varies between users, surfaces and shoes [6], which is why [1] sets its threshold by calibration. Peak detection worked at sampling rates from 20 Hz [5] to 100 Hz [6].

III. Orientation Detection

Heading is obtained in three ways. A magnetometer gives absolute heading once the magnetic vector is rotated into the horizontal plane, since uncompensated tilt causes large errors [5], but indoors the field is distorted by metal furniture, computers and machinery [1], [3], [5]; even a commercial fused sensor is specified at under 1 degree static heading error only in a homogeneous magnetic field [3]. A gyroscope is immune to this but drifts after a few turns [5]; Tian et al. average its yaw over each step and train an offset when the phone does not point along the path [2]. Kang and Han weight the previous estimate, magnetometer and gyroscope by how well the two sensors agree, reaching a 2.28 degree mean heading error [5]; the two sensors disagreed by more than 5 degrees 82% of the time in their indoor tests. Third, when the device orientation on the body is unknown, walking direction is inferred from horizontal acceleration: principal component analysis was the best of five methods in [4], with a 5.7 degree median error, but Jin et al. found it unreliable below 25 Hz and integrated horizontal acceleration over 1.4 s instead [1]. When heading is sampled also matters: orientation is more reliable at foot stance than during swing [3], [5] takes heading at the acceleration valley between steps, and [2] averages it over the step. Filtering angles needs care at the 0/360 degree wrap [1].

IV. Comparison and Implications

Table I summarises the approaches; their accuracy figures are not directly comparable, since each study used different placements, subjects and routes [6]. For a device worn in a fixed orientation, the literature supports peak detection on low-pass filtered acceleration with an amplitude threshold and a minimum step interval, calibrated for the wearer [1], [2], [6]. Placement matters for both tasks: carrying a phone loosely in a handbag caused overcounting [6] and tight trousers degraded the heading estimate in [4], which favours a fixed mounting. Without a gyroscope, heading must come from a tilt-compensated magnetometer [5], so indoor magnetic disturbance is the main expected limitation. None of the six works adopt a magnetometer-only heading; as a baseline in [5] it gave a visibly twisted path, so magnetometer calibration and disturbance rejection deserve the most attention.

TABLE I. Comparison of Reviewed Approaches

Ref.	Sensors, placement	Step detection	Heading
[1]	Acc., compass; two phones in pockets	Vertical acc.; max-min pair, 150 ms gap	Integrated horizontal acc.; device fusion
[2]	Acc., gyro; hand, swinging, pocket	Axis chosen by mode; peak, 0.2 s gap	Gyro yaw averaged per step, trained offset
[3]	Acc., gyro, mag.; foot	Variance of acc. magnitude, two thresholds	Fused IMU orientation at foot stance
[4]	Acc., gyro, mag.; trouser pocket	Variance of vertical acc., threshold	PCA of horizontal acc.
[5]	Acc., gyro, mag.; phone in hand	Vertical acc.; peak, peak-to-valley, slope	Tilt-compensated mag. weighted with gyro
[6]	Acc.; six phone placements	Acc. magnitude; windowed peak detection	Not covered

References

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